

Standardised Test Cohorts for Evaluation

Without standardised cohorts, different models (e.g. sequence vs. tree-based) would be evaluated on slightly different samples, leading to unfair or misleading comparisons. To ensure consistent and fair comparison across all models (LSTM, TCN, XGBoost, LightGBM, EVL), we constructed **four standardised test cohorts** from the NSW demand dataset. These cohorts align all models to the same timestamps and ground truth values, eliminating discrepancies caused by lookback trimming, forecast availability, or sub-sampling.

The raw test split (df_test) contains **38,737 rows** (half-hourly timestamps). However, models trained with a **48-slot lookback window** cannot use the first 48 slots of the test set, since lagged features are undefined at the start. This trimming produces **38,642 valid supervised examples**.

Now for the 24h-matching cohorts:

- overall 24h-match = 25,760 ($\approx$ **67%** of total test samples).
- hot-day 24h-match = 2,560 ($\approx$ **67%** of hot-day samples).

cohort_overall_all.csv	• All valid supervised test samples after applying the 48-slot lookback. • Used for pure model-to-model comparisons without requiring 24h forecasts. • Size: 38,642 rows.
cohort_hotday_all.csv	• Subset of overall_all, restricted to is_hot_day = 1. • Evaluates model behaviour under extreme heat days using the full test set. • Size: 3,840 rows.
cohort_overall_24match.csv	• All test samples where AEMO's 24h forecast (y_forecast_24h) is available. • Ensures fair comparison between models and the 24-hour forecast baseline. • Size: 25,760 rows.
cohort_hotday_24match.csv	• Subset of overall_24match, restricted to samples where is_hot_day = 1. • Used to analyse extreme-day behaviour when 24h forecasts exist. • Size: 2,560 rows.

Each file includes:

- timestamp (aligned test datetime)
- is_hot_day (0/1 flag)
- true (ground truth demand 24h ahead, MW)

- op_24h (AEMO's 24h forecast, MW; NaN where unavailable)
- fs_strong (prediction of the best LSTM model, MW)
- hot_day_a15 (prediction of the fs_strong tuned with hot-aware weight $\alpha=1.5$, MW)

This structure guarantees that all participants use identical targets, baselines, and timestamps for evaluation.

Usage

- **Overall performance:** use cohort_overall_all.csv to compare across all test samples.
- **Extreme-day performance:** use cohort_hotday_all.csv to isolate hot-day errors.
- **Forecast baseline comparison:** use the *_24match cohorts to compare models directly against AEMO's 24-hour forecast (y_forecast_24h).
- For fairness, all error metrics (MAE, RMSE, MAPE, Bias) should be computed on these cohorts only.